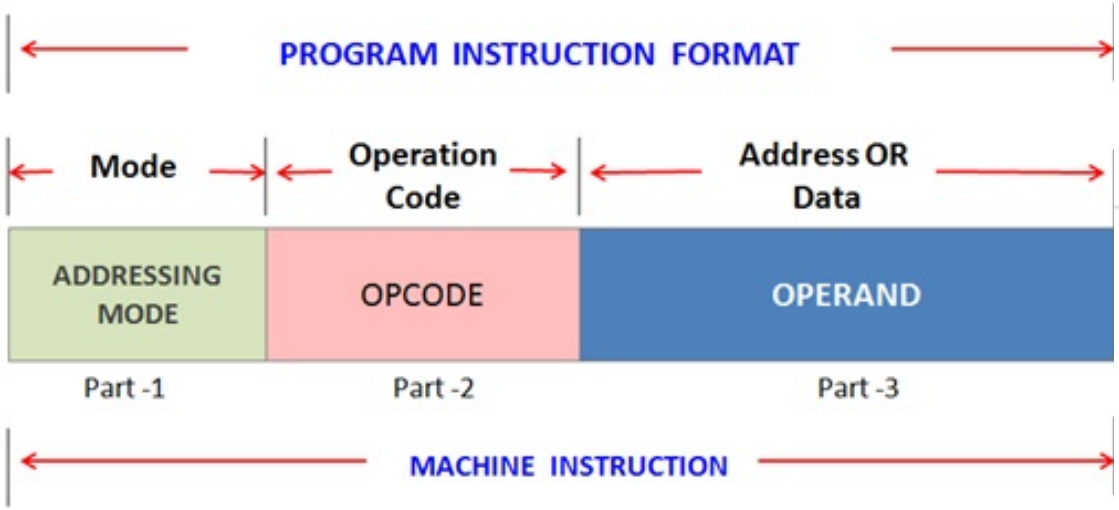
Addressing Modes

CSE Sem-4

Instruction Format

- Instruction format is a pattern of bit, that control unit can decode.
- Each field in Instruction format specifies a particular information to the processor

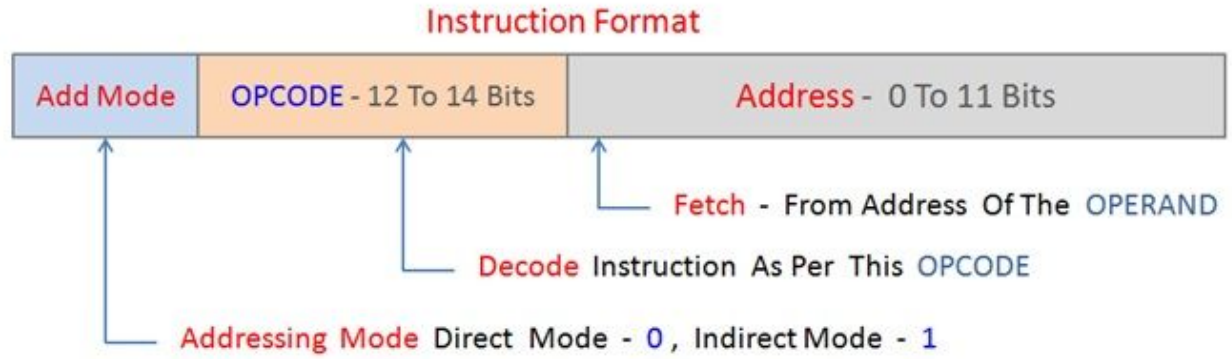


PART 1 – Addressing Mode - Rule For Operand - Data Or Address

PART 2 – OPCODE - For Control Unit - Which Operation To Perform.

PART 3 – OPERAND - For ALU - On Data Operation To Be Performed.





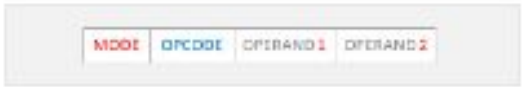
- Value of addressing mode:

  0 indicates direct referencing. Operand contains data.

  1 indicates indirect referencing. Operand contains address of data

- Direct Address Mode.
- Indirect Address Mode.
- Implied Mode.
- Immediate Mode.
- Register Mode.
- Register Indirect Mode.
- Auto Increment Mode.
- Relative Address Mode.
- Indexed Addressing Mode.
- Base Register Mode.

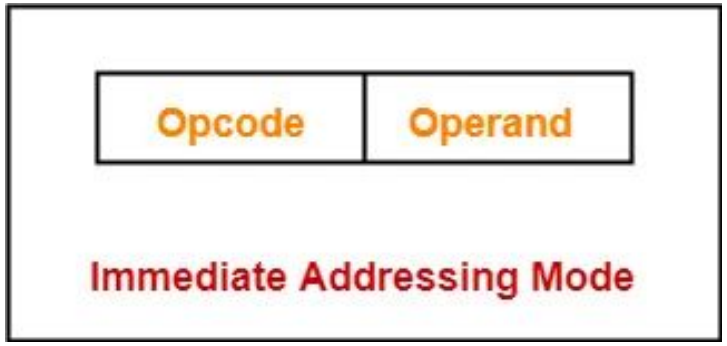
Instruction Type

Instruction Format Type	Instruction Format	Example	
Zero Address Instruction		CMA	Instruction format without any operand. Used in stack.
One Address Instruction		ADD 06H LDA 20H	Instruction with only one operand. Use accumulator to store other operand and result.
Two Address Instruction		MOV R1 , R2 ADD AX , BX	Instruction format with two operands.
Three Address Instruction		ADD R1 , R2 , R3 SUB R1 , R2 , R3	Instruction format with three operands.

1. Immediate Addressing Mode

- The operand is specified in the instruction explicitly.
- Instead of address field, an operand field is present that contains the operand.

```
ADD R1  #20
MVI B   #1
```



- As constant value is same throughout the program, so in instruction, without giving address of operand, directly the value is given.

Disadvantage:

Range of operand/constant depends on the length of the address field of the instruction. If constant size is greater than the address field, the length of the instruction will also be increased.

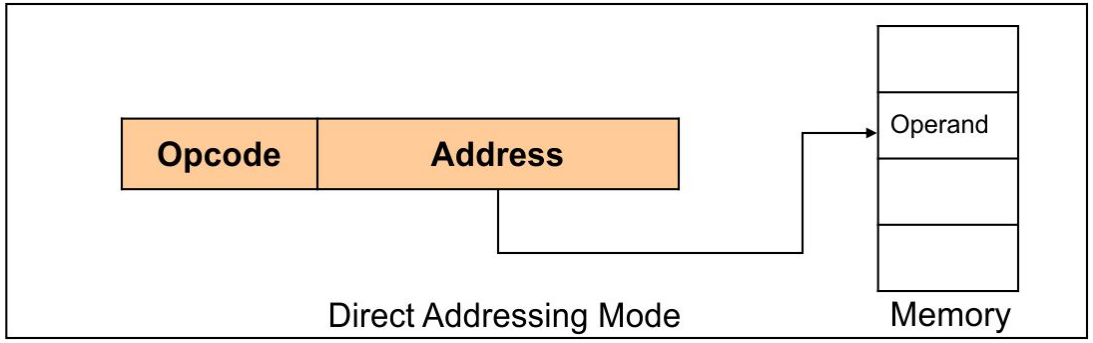
For example, for a 16-bit instruction, 1 bit is used for addressing mode, 3 bits are used for the opcode. And remaining 12 bits are used to specify the constant. So constant value size can not be greater than (2^12=) 4096. Therefore, range of constant will be 0 - 4095

2. Absolute/Direct Addressing Mode

- Operand field contains actual memory address of the operand.
- Use to access variables.
- Effective Address (EA) = address field of operand

ADD A, 400H
⇒ Acc = Acc + M[400]

LOAD R1, 1000H
⇒ R1 = M[1000]



Advantage:

✔ No computation required to calculate the effective address.

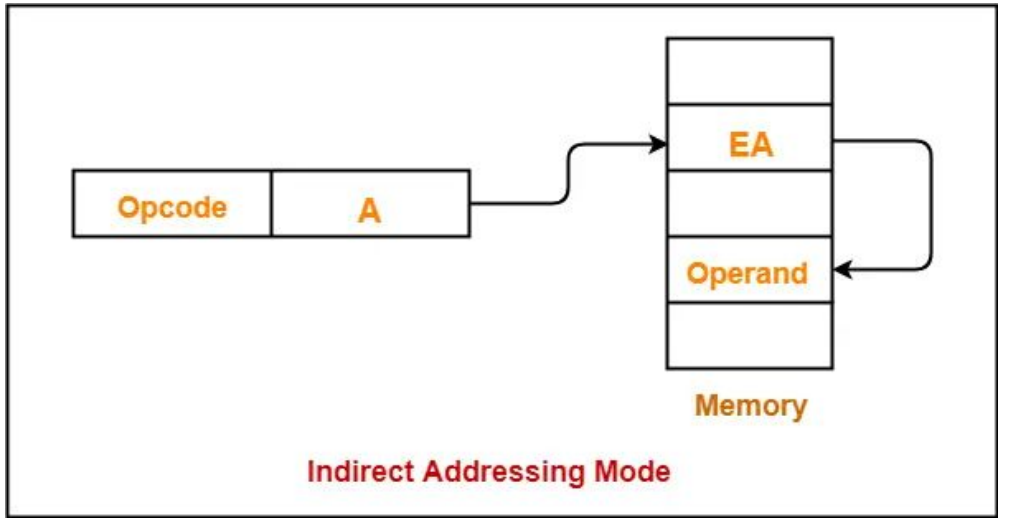
Disadvantage:

× If instruction size is fixed to 16 bit, 12bits will be available to store the address of a memory location. So, we cannot give the address of operand greater than 12 bits. In other words, only 2^12=4096. (range 0 – 4095) memory locations can be fetched through it.

× If number of memory blocks increases, instruction size will also get increased.

3. Memory Indirect Addressing Mode

- The address field of the instruction contains the address of memory location. That memory location contains the effective address of the required operand.
- Two references of memory are required to fetch the required operand.
- Used to access pointers or passing parameters.



Indirect Addressing Mode

LOAD R1, 1000H

⇒ R1 will be loaded with the content, whose address is stored in memory location 1000H

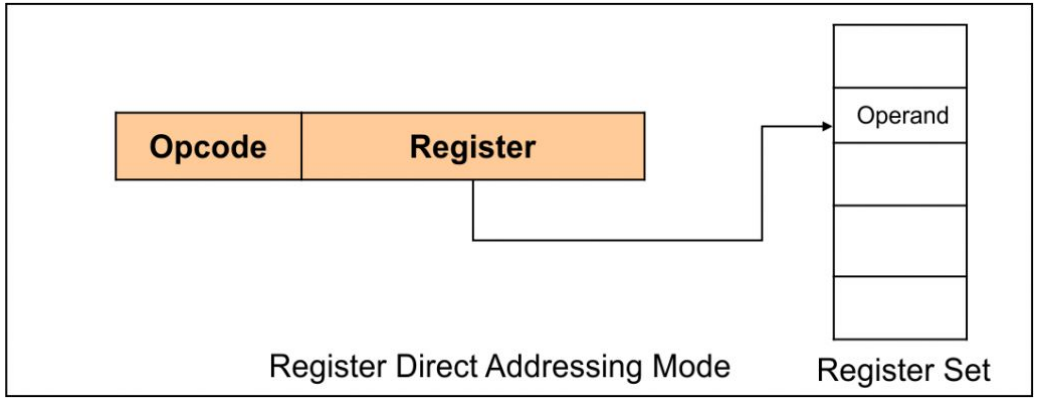
ADD X

⇒ Acc = Acc + M[M[x]]

Effective Address (EA) = M[x]

4. Register Direct Addressing Mode

- The operand field of the instruction refers to a CPU register number.
- That register contains the operand.
- In this mode operand is stored in the register.
- No reference to memory is required to fetch the operand.



ADD R3, R4

⇒ R3 = R3 + R4

Advantage:

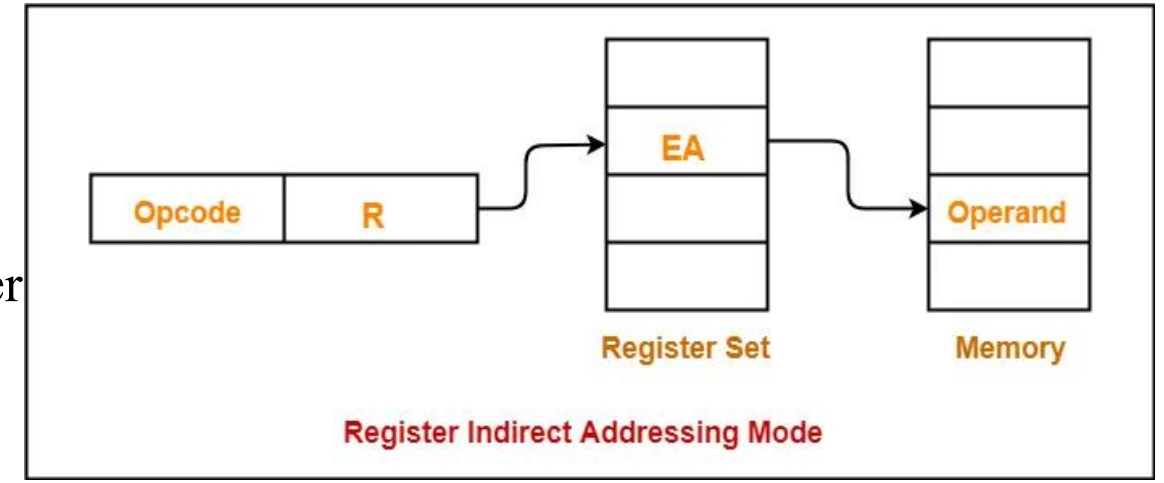
✔ Instruction size is small, as no of registers in processor is limited.

Disadvantage:

× It has limited address space

5. Register Indirect Addressing Mode

- The operand field of the instruction refers to a CPU register number.
- That register contains the memory address of the operand.
- In this mode operand is stored in the memory. And a register contains the address of that memory location.



ADD R3, R4

⇒ R3 = R3 + M[R4]

Effective address = Content of register specified in the instruction

Advantage:

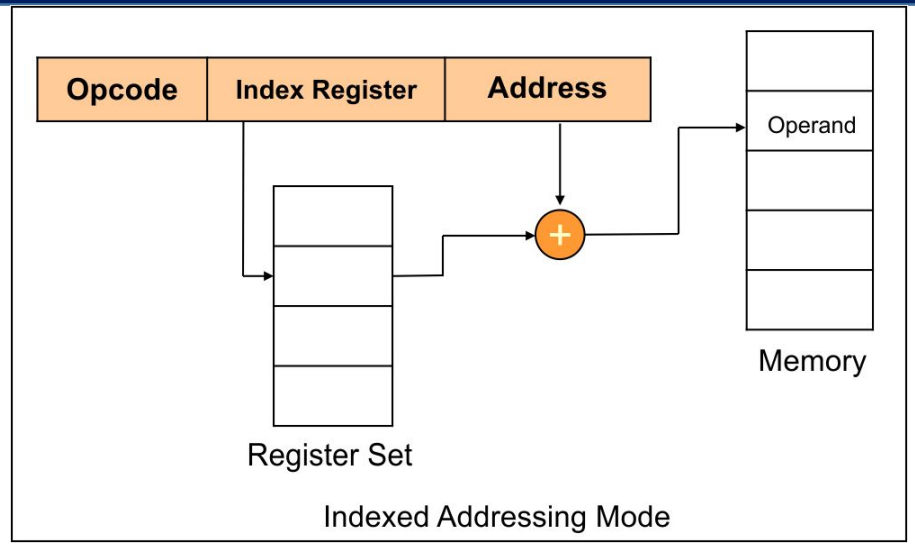
✔ Instruction size is small, as no of registers in processor is limited. So even to access a memory location, a smaller number of bits must be included in the operand part of instruction field.

Disadvantage:

× More computation required to find the operand.

6. Index Addressing Mode

- Use to access arrays
- Operand-field contains the starting or base address of the array-memory block.
- Index-register field will contain the index value, means displacement.



LOAD R2, 4(R3)

⇒ R2 will be loaded with the content present at the location obtained by adding 4, i.e. displacement (content of indexed register), with the starting address of the array (base address) stored in R3

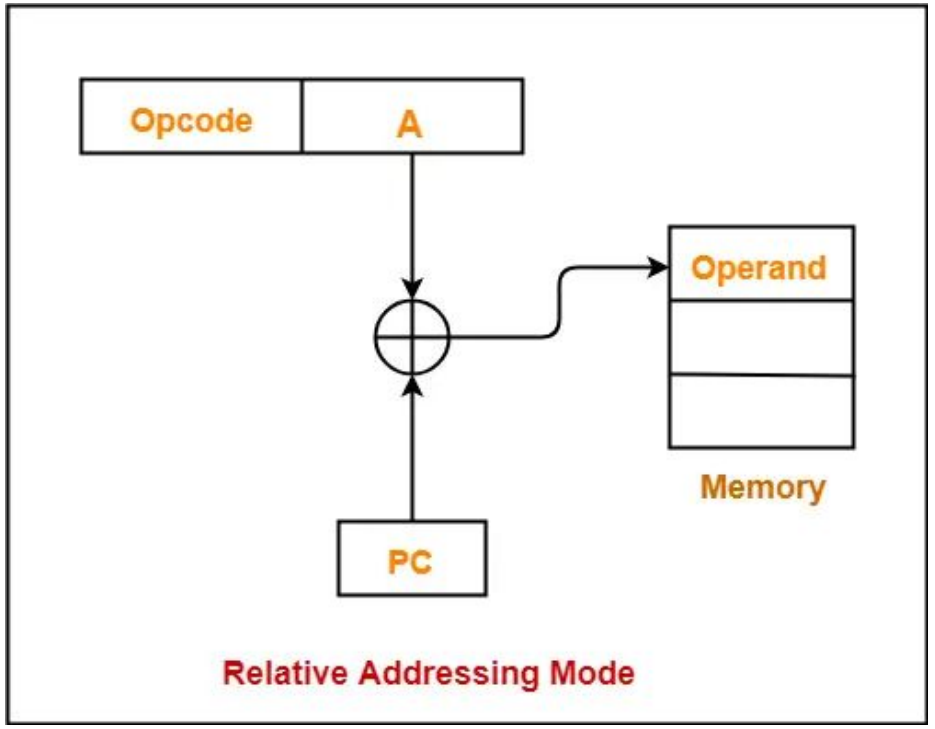
Effective address = Base address + Index register

Advantage:

✔ Instruction will not be changed for fetching the next array element. Only value of index register will get changed.

7. Relative Addressing Mode

- ❖ Used for branch, jump or control instructions
- ❖ Operand-field contains the displacement.
- ❖ Effective address of the operand is obtained by adding the content of program counter with the address part of the instruction.



BR 49

⇒ If the program execution is at 500H locations in main memory and we have a branch instruction as BR 49. It means going to 550H location after 500H.

Effective address = PC + Displacement

Note:

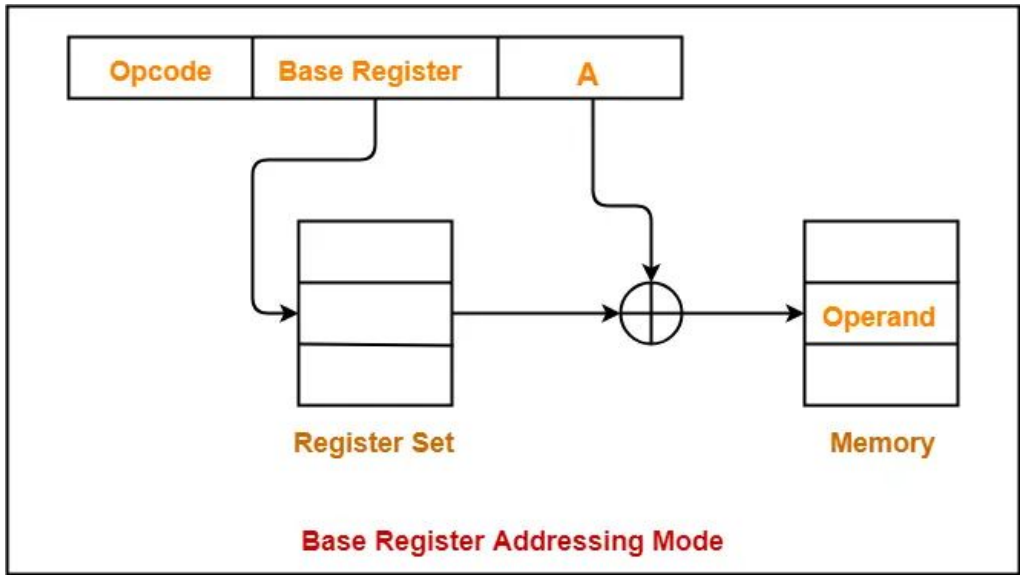
- Program counter (PC) always contains the address of the next instruction to be executed.
- After fetching the address of the instruction, the value of program counter immediately increases.
- The value increases irrespective of whether the fetched instruction has completely executed or not. So, displacement value should be 1 less than the original displacement.

Advantage:

✔ Use of offset, reduce the instruction size. Because we give offset value in instruction rather than actual address.

8. Base Register Addressing Mode

- Used in program relocation
- The effective address of the operand can be found by adding the value of the base register with the address part of the instruction.
- If cache memory is full, swap a program out from memory and new programs are swap-in. After some time again load the same swap out program in cache memory with the same or different location.
- Base register store the starting address of a program. And for branch instruction, displacement is stored in the address field.



Effective Address = Content of Base Register + Address part of the instruction (Displacement)

Note:

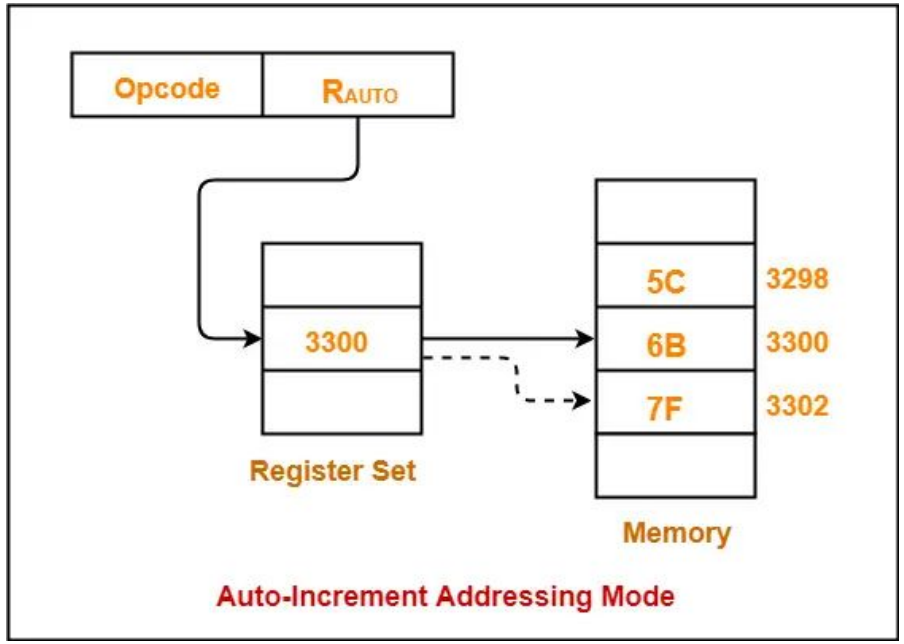
- Relative and based register both addressing modes are suitable for program relocation at runtime.
- Based register addressing mode is best suitable to write position independent codes.

Advantage:

✔ Reduce the number bits in the addressing field of the Instruction

9 (a). Auto Increment Addressing Mode

- Effective Address of the Operand = Content of Register
- After accessing the operand, contents of this register are automatically incremented to point to the next consecutive memory location.
- Useful for stepping through arrays in a loop.

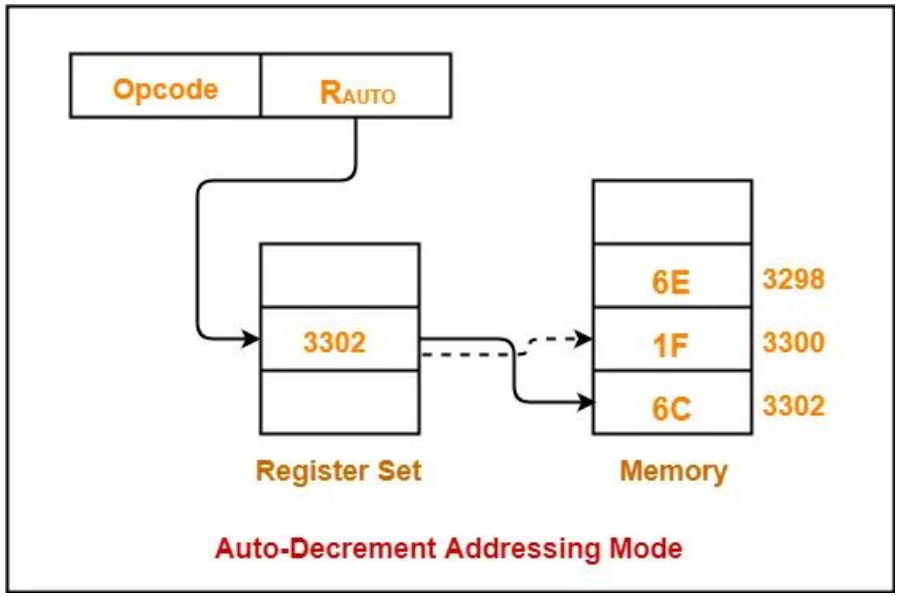


ADD R1, (R2)+
R1 = R1 + M[R2]
R2 = R2 + d

⇒ First, the operand value is fetched. Then, the instruction register value is incremented by step size ‘d’.

9 (b). Auto Decrement Addressing Mode

- Effective Address of the Operand = Content of Register
- Before accessing the operand, contents of this register are automatically decremented to point to the previous consecutive memory location.
- Useful for stepping through arrays in a loop.



ADD R1, - (R2)
R2 = R2-d
R1 = R1 + M[R2]

⇒ First, the instruction register value is decremented by step size ‘d’. Then, the operand value is fetched.

10. Implied Addressing Mode

- The operand is specified in the instruction itself.
- It is also called as implicit addressing mode.
- It is mainly used for Zero-address (STACK-organized) instructions. For such instruction, the operand is given in the instruction. We have no need to find the operand.

CLA => Clear Accumulator
CMA => Complement Accumulator

